Guide to the research lifecycle in DP-Next

Luke W. Johnston

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Welcome!

> **Note**
>
> This guide mostly follows the diátaxis "how-to guide" style, though not strictly. It is a living and constantly evolving guide that is regularly updated as we learn and refine how we work and develop data packages. We intend to continually update and release every update to the guide to Zenodo and as GitHub releases. We don't expect this guide to ever be considered "done".

There are surprisingly few, practical resources and how-to guides for learning about and doing effective collaboration within research, especially in the context of open science and reproducibility. The few resources that do exist are more theoretical than the practice "how do I actually do it?" aspects.

In particular, few research groups and institutions develop and curate their own guide on how they do things. This makes it very difficult to learn as a new (or established) researcher how others do things, so that we can draw inspiration from or build off of how others work and do research. Often these skills are gained through verbal knowledge sharing or from watching how peers or other group members work. This makes it difficult to apply any type of scientific reasoning, critical thinking, and analysis to these collaborative, open scientific, and reproducible practices.

This guide is an attempt at documenting and sharing how we aspire to and aim to work within the DP-Next project.

Who you are

We've written these documents with a few types of people in mind:

- **DP-Next collaborators/group members**: These people are our primary audience for this guide. If you work with us in DP-Next, these documents are designed and developed with you in mind.
- **Researchers not affiliated with DP-Next**: If you work in another organization or group, we write these documents to share our practices with you. You can use this either as a guide or as a reference to help you improve your own work or to draw inspiration from for how you structure your research and collaboration.
- **Group or organisation leaders**: If you are a leader of a group or institution, we write this guide to show how a group of researchers *could* work together in more effective ways. You can use this guide to get inspiration for how you can design your own group or institution to work better together at producing open and reproducible research.

Contributing

Check out our contributing document for information on how to contribute to this guide.

Changes

Check out our changelog for information on what has changed in each version of the guide. Previous versions can be found in the releases page of the GitHub repository.

How the website is made

The website and PDF (download link on the top of the website's sidebar) are created using Quarto to write the material and create the book format. GitHub hosts the Git repository of the material, while GitHub Actions builds and releases the book with every change. Netlify hosts the website and manages the domain. The source material is in the `guide` GitHub repository.

Inspirations

- The Turing Way

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1. Overview

2. Practicalities

This section contains some higher-level, practical things that relate to many aspects of working in DP-Next.

2.1 Work as openly as possible

As much as possible, work on public GitHub repositories and upload your work as openly as possible. There are few things that truly must be private, but for those that are, use private GitHub repositories.

2.2 Do all work on GitHub and/or Git, including managing projects

All work (that can legally be) needs to be done on our DP-Next GitHub organization. This provides a central place that we can do work on together, makes it easier to keep work visible to others in the group, and makes it easier to resolve issues and help each other out. It also makes it easier to do work more openly.

Use GitHub Project boards to follow a Kanban-style approach to project management. Use GitHub issues to track tasks, write comments and have discussions in issues. That ensures any relevant discussions or conversations happen within the same space as the project, so it's easier to find later.

Create pull requests when making changes to files when you want a review of those changes. Add review comments directly in the pull request, so that the comments are directly connected to the changes being made and can be more easily resolved in the same space.

2.3 Write documents with Quarto Markdown

Markdown is a way of writing plain-text that is open, easy to work with for all operating systems (e.g. Mac, Windows, Linux). It uses simple syntax to bold words (e.g. `**bold**`) or make headers (e.g. `# Header`). We write all documents in Markdown, specifically Quarto Markdown. Quarto is very powerful, as you can mix R code and text in Quarto, making it easier to create reproducible analyses and papers. You can also easily create websites, PDFs, and other output from it. Quarto (or Markdown) files have the `.qmd` (Quarto Markdown) or `.md` (Markdown) file extension.

2.4 Process and analyze data with R or Python

R or Python are open source languages that are commonly used in research and science. Read our decision post on why we use R or Python. It's also easy to use R or Python in Quarto to write reproducible scientific outputs like papers. That means you can use R or Python code to generate your figures and tables for you and automatically add them to your paper or document.

2.5 Name files following our style convention

How files and folders are named is one of the most effective ways of improving collaboration and making it easier to work on and maintain a project. As much as possible, name files using `kebab-case` (lowercase and with `-` for spaces). For example, `paper.qmd` or `protocol.qmd`. Some files are capitalized by convention, e.g. `README.md`.

Why not use spaces in filenames? Spaces can make things difficult to work with for some systems and programs. For instance, if we want to generate our files into a website, including spaces in the names can make it difficult to generate the URLs for the pages.

Name files simply and as short as is reasonable, using as fewer words as is appropriate or even better, use single words. Prefer nouns rather than verbs. Avoid special characters, including Danish characters like `æ`, `ø`, and `å` as these can cause some problems on some systems. Also avoid repeating the same word in the filename that would also in the filepath. For example, instead of `contributing/`

`contributing-next-steps.qmd` use `contributing/next-steps.qmd` or instead of `project-name/docs/dp-next-project-name-paper.qmd` use `project-name/docs/paper.qmd`. The filepath describes the contents of the file, not just the filename itself.

2.6 Communicate with others using Discord

We use Discord for general communication and coordination in DP-Next. Why Discord? Check out Seedcase's decision post for details on why. In general, we use Discord for more informal communication, including for video meetings. For project-specific communications, it's best to create an issue or comment in an existing issue in the relevant project.

2.7 Keep your `dp-next.yaml` file up-to-date

Throughout the project life cycle (also after the manuscript / conference abstract is submitted), you need to keep the status of the project updated in the `dp-next.yaml` file. We use this file to manage projects, keep an overview of activities in DP-Next, and to make it easier to display that information on the main website.

2.8 Publish nontraditional components of research (in Zenodo)

Part of the aim of DP-Next is to publish nontraditional research outputs, not just papers in journals. For example, we publish protocols, analysis code in GitHub repositories, and preprints. We publish these outputs on Zenodo, mainly because it has a great programmatic access to uploading outputs there, it's based in EU, and is nonprofit. Seedcase also has the zen-do package that makes it much easier to automate publishing to Zenodo. Using Zenodo means we get a Digital Object Identifier (DOI) and allows us and others to cite these works.

3. Roles and responsibilities

In a larger project there are many tasks and needs to be done, many more than any individual person can handle at any one time. To manage this, we describe specific responsibilities that need to be done within DP-Next and assign them to people so that people's roles are clear and they know what they are responsible for.

3.1 Roadmap management

A roadmap is a set of products, milestones, and timelines necessary to complete a particular project. In our case, a project would be an individual work package.

An effectively developed and regularly maintained roadmap is critical for successful and timely completion of a project. But they aren't just for planning purposes, they also are extremely effective at communicating the project's status, next steps, progress, and direction to everyone involved in the project. It keeps everyone aligned and makes it clearer what needs to be done.

Because roadmaps are so important and require regular maintenance and management, someone needs to be responsible for them. The roadmap manager's tasks are:

- Identify all concrete products for a project (one product is one repository) by talking to the collaborators and list them in the roadmap (see how below). This is a critical step, but also can be extremely difficult to do, as knowing clearly what a product is and what products are needed for the project can be hard for collaborators to themselves communicate and agree on. This is part of the responsibility of the roadmap manager to dig in and ask detailed and probing questions. This will be an iterative process, so don't expect to get clear answers right away.
- Identify all milestones for each product, again by talking to the collaborators by probing and digging in to get as clear an idea and picture as possible on what needs to be done for each product. Then list them in the roadmap. Like the products, this is something that will need to be done regularly and will evolve over time. So expect to regularly talk with collaborators about milestones and tasks.

We use GitHub Projects to create and manage roadmaps (see some of our roadmaps in our list of project boards). These are some things that we do in the GitHub Projects to manage our roadmaps:

- Use the roadmap view in GitHub Projects as the main view for the roadmap project board. Group this view by "product" (see below) to see the milestones (see below) grouped by product. This main view should be filtered by `-status:Done`, so that only active milestones are shown.
- Create draft issues and use them to represent milestones for a given product. Start and end dates should be put on each milestone.
- Create a custom, single-select field to represent the project's products and within that field, add options for each of the products in the project. One product should represent one repository. Each milestone ("draft issue") should be assigned to a product using this single-select "products" field, so that milestones are grouped by product in the roadmap view.
- When a milestone is complete, change the status to "Done" so it gets filtered from the main roadmap view.

3.2 Issue management

GitHub Issues are the primary way we track work in DP-Next. We use issues to describe specific tasks that need to be done, to have a place to discuss and brainstorm about particular topics, and come to an agreement or decision for those topics. As a product progresses, more issues are made, often faster than they can be completed. And issues can quickly accumulate, go out-of-date, or be forgotten. Combined with the fact that the priority of any given issue needs to reflect the current state of the roadmap, someone needs to be responsible for curating and managing the issues for a given product. The issue manager's tasks are:

- Understand the product enough to create issues that contribute to its completion.

- Track and ensure issues are completed and closed, by reminding assignees of their issues and checking on the status of issues regularly (ideally during update meetings)
- Close any issues that are no longer relevant or are out-of-date.
- Consult the roadmap to identify what issues need to be prioritized in any given period of time.
- Ensure that the distribution of the issues priorities are balanced. If all issues are high priority, no issue is a high priority. There needs to be a balance between high, medium, and low priority as it reflects an understanding the actual state and needs of the product and overall project.

We use GitHub Issues as well as GitHub Projects to manage our issues. These are some things that we do in GitHub to manage our issues:

- Use a Board view in GitHub Projects, grouped by "status" to show "todo", "in progress", and "done" columns. This is the main view for managing issues. Create "swimlanes" on the board by assignee so each person has their own "lane" to track and manage their own issues.
- Create a custom, single-select field to represent the issue's priority and within that field, add options for "high", "medium", and "low" priority.
- Assign priorities to issues using the "priority" field. An easy way to do this is to create a new view (just for the issue manager) that has all the issues grouped by priority. With the grouping, it is easy to drag issues between the different priority levels to assign them. This also makes it easy to see the distribution of priorities across all issues and ensure that they are balanced.
- Keep the main view (grouping by status and swimlane by assignee) filtered by `has:priority`, so that only issues with priorities are shown. This helps keep the focus on and communicates to everyone what actually needs to get done.

4. Project management

Every task that each person is working on is created, tracked, and visible on the GitHub Project board. The board is divided into columns that represent the state of the task (e.g. “to do”, “in progress”, “in review”, “done”). The board is used to track the progress of the project and to identify bottlenecks or issues that need to be addressed, reviewed, or discussed. We also assign tasks to people on the board, which is shown on the board itself to make it clear who is working on what.

All discussions happen on the GitHub issues, which are linked to the tasks on the board. These discussions are comments to the Issue, that way we keep all the information about a task in one place and so that others can participate in the discussion as well.

5. Meetings

5.1 Check-ins and "retrospectives"

Every few months we have a "collaboration-assessment" meeting. At the least, the tech leads and/or representatives from each work package need to attend, but it is open to anyone working on DP-Next. We follow the style of a meeting type called a "retrospective" that is also used in the Seedcase Project.

During these meetings we discuss any challenges people are having and ways that we can improve collaboration within and between work packages. The focus is mainly on collaboration practices, tools, and methods that could benefit the project as a whole or multiple work packages. We discuss what went well, what didn't go well, and what we can improve. Then we create action items to address the issues we identified.

5.2 Update meetings

Every week, most work package groups have a very short "update" meeting. During the meeting, the roadmap manager shares their screen of the project board and go over the tasks on the board that are at the "in-progress" or later (e.g. "in review") states. Each person gives an a very brief update on their tasks they've been working on since the last update meeting. Pay attention to tasks that are blocked and briefly discuss if any action needs to be taken to resolve the block.

> Tip
>
> This style of meeting is modelled after the Seedcase Project update meetings, which itself models these meetings off of "daily standups" and similar iterative and incremental project management practices.

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6. Getting started

7. Start a new project

If you are part of DP-Next (or not), you'll be working on research projects, likely using data collected during DP-Next or from existing data (e.g. ADDITION-PRO). This chapter describes what to do when starting a new project. Below in Figure 7.1 is a visual summary of the steps to take, along with descriptions afterwards that outline the steps in more detail.

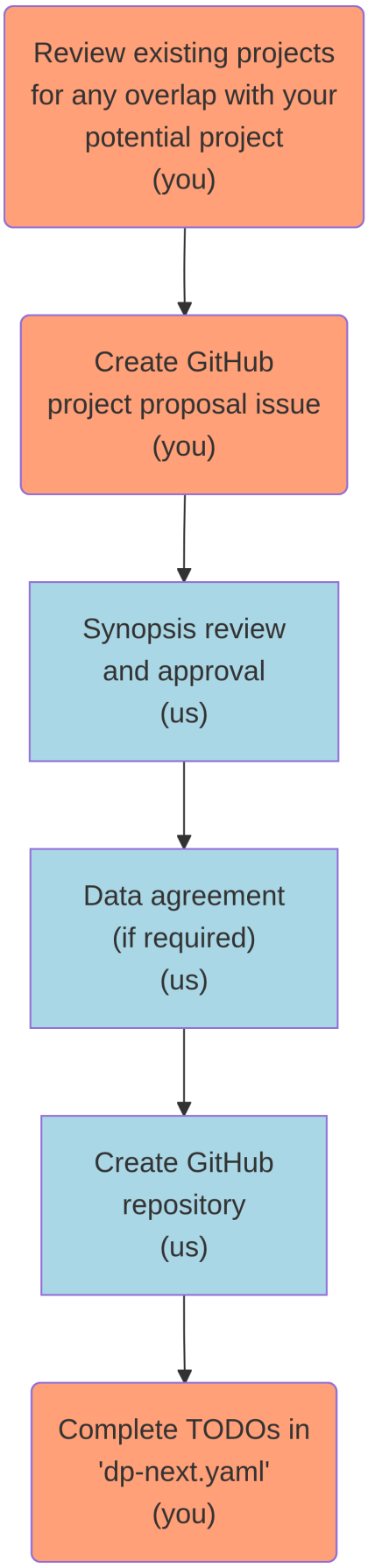


Figure 7.1: Steps to take for starting a project within DP-Next. The rounded boxes in orange are tasks you are responsible for doing. The rectangular ones in light blue are for the managers ("us") to do.

The requirements for preparing a DP-Next publication apply to both internal and external researchers and should follow these steps:

1. Before proposing a project, make sure that you aren't doing something that's already done or being worked on. If there is a current ongoing project that is similar to what you wanted to work on, you can request to collaborate on that project with those researchers.
2. If there is no other project that is similar to your project idea, you can propose the project. Create a new issue in the WP1-Management repo (private for now) and select the "Project proposal" issue template. Fill in the fields there, which we'll use to assess how relevant the project is within DP-Next.
3. Once you've created the issue, at least one member of the Steering Committee needs to review and approve it (usually either the principle investigator or the general coordinator). It is possible that someone may approve it, but later another committee member may request changes or give comments. If that happens, it is fairly easy to update the project. Once approved, someone with access permissions (e.g. in WP1) will create a GitHub repository for the project by using the DP-Next template project and following our naming convention.
4. After the repository is created, you need to go through the repository and complete any TODO items, such as those found in the `TODO.md` file. Importantly, the TODO items in the `dp-next.yaml` file need to be filled. We use the information in this file to create an overview of the different projects going on in DP-Next.

> **Note**
>
> We expect that many new projects based on DP-Next data will be done by DP-Next researchers. However, we also welcome collaborations with external researchers (i.e. those who are not part of DP-Next), though these projects must include at least one DP-Next member as coauthor so that projects follow our standards. Projects can be stand-alone or part of a wider initiative, such as a PhD or postdoc protocol.

> **Warning**
>
> For investigators not covered by the DP-Next agreements (by institutional affiliation to one of the Steno Diabetes Centres), a separate data agreement must be signed if they require data access. This agreement must be in place before any data access can be given.

8. Denmark Statistics

Working on the Denmark Statistics (DST) servers brings with it unique challenges when it comes to collaborating together. This page describes how we've set it up to effectively collaborate on the DST servers.

> **Note**
>
> We show the steps to take using the terminal. Why? Because a lot of power and effectiveness comes from using the terminal and some steps can only be done in the terminal. It is also just easier to show a series of commands to run in the terminal rather describe how to do it with R, which requires more steps to take (e.g. opening up RStudio and finding the right working directory).

8.1 Configure Git

The very first step is to configure your Git. You thankfully only need to do once (or after DST has reset your profile). If you haven't already configured your Git, open a terminal (called Git Bash in DST) and type out these commands:

```
git config --global user.name "YOUR NAME"
git config --global user.email "name@email.com"
git config --global init.defaultBranch main
```

Replacing the `YOUR NAME` and `name@email.com` with your actual details.

To check whether it worked, you can run this command:

```
git config list --global
# Or if that doesn't work
git config --list
```

In the output, you should see the configurations you've set.

8.2 The folder structure

Before setting up this folder structure, the first thing to decide is what the "product"/output will be and what its name (or abbreviation) will be. The reason for this, as per our design principles, is that we want to structure work around products and around teams (or groups), rather than individual people. So a folder structure should reflect how to work on the product as a group.

We'll use an example of a product called `wp2-analysis` that will be done within DST. There are two people working on this product (for now) called `person-name-1` and `person-name-2`. The folder structure will look like this (don't create it yet!):

```
workdata/<project-id>/
└── wp2-analysis/
    ├── remote/ # Contains only Git specific folders
    ├── person-name-1/
    │   ├── .git/
    │   ├── data/
    │   ├── docs/
    │   ├── R/
    │   ├── wp2-analysis.Rproj
    │   ├── DESCRIPTION
    │   ├── _targets.R
    │   └── README.md
```

```
    └── person-name-2/
        ├── .git/
        ├── data/
        ├── docs/
        ├── R/
        ├── wp2-analysis.Rproj
        ├── DESCRIPTION
        ├── _targets.R
        └── README.md
```

The `remote/` folder is a (bare) Git repository, meaning that the only files and folders found within it are Git specific folders and files, such as `branches/`, `hooks/`, `objects/`, `refs/`, `config`, `description`, and `HEAD`. The other folders (for each collaborator) are the "local" repositories that have the standard structure for data analysis projects. In this case, the "remote" folder is a bit like a GitHub repository, while each collaborator's "local" repository is like their own local repository of the GitHub (remote) repository.

> ! Important
>
> We'll describe the steps later on, but an important note is that each individual person will need to create their own folder. That's because servers like DST assigns the "owner" of the folder to the user who created it, so you may encounter issues if someone else made the folder for you.

8.3 Creating the "parent" folder

You need to first open a terminal in the right location and create the correct folder. To do that, open up the project's `workdata/` folder using the File Explorer in the DST Windows desktop. The project folder should be at a location like `E:/workdata/7000000/` (the project ID is usually a seven-digit number starting with 7). When in this folder, right-click somewhere in the folder and select the menu option "Open in Git Bash" (or something similar). A terminal program should open up at this location. In the new terminal app, create the new research project folder (`wp2-analysis` from the example above):

```
mkdir wp2-analysis
```

The `mkdir` command means "make directory" (or folder). Next, you want to move your "working directory" (the folder location your terminal is in) into this newly created folder using the `cd` command (change directory):

```
cd wp2-analysis
```

You are now ready for the next steps.

8.4 Creating a Git "remote" as a folder

To create the "remote" folder, first create the `wp2-analysis/` folder. Then, open a terminal into the `wp2-analysis/` folder and run the following command:

```
git init --bare remote
```

This will then create the `remote/` folder with the necessary Git files and folders that will allow it to be treated as a Git remote repository (like with GitHub). This is the folder each collaborator will push and pull to (more on that later below).

8.5 Creating the first "local" repository

Next, one of the collaborators (e.g. `person-name-1`) needs to create their local repository, ideally using a standard structure as set up by the prodigenr package. To do this run the following R function while in the terminal:

```
Rscript -e 'prodigenr::setup_project("wp2-analysis")'
```

This will create a folder with the `.Rproj` file name as `wp2-analysis`. But since we want each subfolder to be the name of the collaborator, you'll need to rename the folder to `person-name-1` (name or other identifier of the person doing these tasks, without spaces). You can use the `mv` command in the Git Bash terminal, which renames a file or folder to a new name using the pattern `mv OLD NEW`.

```
mv wp2-analysis person-name-1
```

Next, you'll need to connect this local repository to the `remote/` folder. Before you do that, you'll need to “move into” that new folder by using the “change directory” command `cd`.

```
cd person-name-1
```

Next, you can add all the files in that directory into the Git history:

```
git add .
git commit -m "start of project"
```

Now that this folder has files saved to the Git history, you can connect the repository to the `remote/` folder.

```
git remote add origin ../remote
```

You'll use the relative path `../remote` to mean “one folder up” (`../`) as you will always be keeping both the remote and “local” (collaborator) repositories within the same parent folder (e.g. `wp2-analysis/`).

Next you'll need to push the Git history to the remote, while also telling the project's Git to use that remote whenever you use push and pull with the `-u` option:

```
git push -u origin main
```

You're now ready to collaborate! :tada:

8.6 Creating more “local” repositories

Now that the “remote” folder has been officially set up, it's easy to add another “local” repository for another collaborator (e.g. `person-name-2`). Replace `person-name-2` with the name (or other identifier) of the second collaborator (without spaces). This step should be done by the other collaborator (e.g. the person who's name is `person-name-2`). You'll need to open the terminal in the `wp2-analysis` folder. You can check if you are in the right folder by using the `pwd` command (print working directory).

```
pwd
```

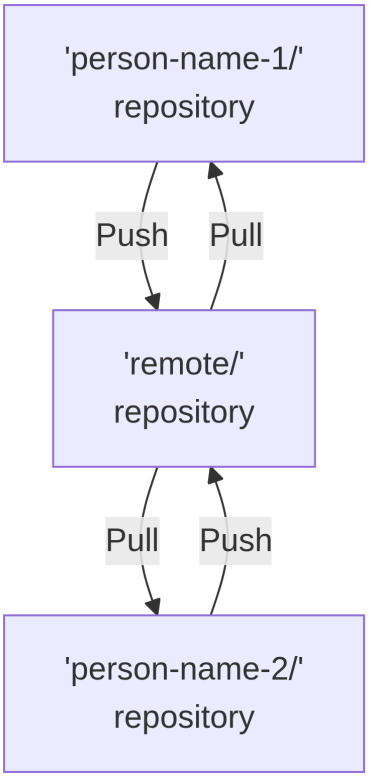
And it should output: `E:/workdata/PROJECT-ID/wp2-analysis`. If you are *not* at that location, make sure to open the File Explorer, find that folder, and right-click somewhere to select “Open in Git Bash” in the menu. When you're in the right folder, run this command:

```
git clone ./remote person-name-2
```

This will create a new folder called `person-name-2/` with the same structure as the first collaborator's local repository. It will also be connected to the same `remote/` folder for pushing and pulling changes. This collaborator can now open the project in RStudio, Positron, or any other editor they like.

8.7 Using the “remote” to collaborate

Now that you have the remote and multiple local folders for each collaborator, you can start collaborating together! Whenever either collaborator makes changes to their local repository, regularly “push” and “pull” to the remote (via RStudio's interface or via the terminal) so that the remote is always up to date. This can be visually represented as follows:



This way, both collaborators can work on their own local repositories, but they can easily share their changes with each other by pushing and pulling to the remote repository. It also means that any collaborator can contribute to the product repository without impacting the work of other collaborators. This allows for effective collaboration while working on the DST servers, even without access to external platforms like GitHub.

There will of course be some issues that will come up, especially merge conflicts. If each collaborator is pushing and pulling to the `main` (could also be called `master`) branch, then you will likely encounter merge conflicts. This isn't necessarily a bad thing, as you can always resolve them. And it's also fairly easy to avoid if you communicate with each other and coordinate your work effectively, e.g. by saying "hey I'm working on this file".

A more powerful approach is to use branches for each change. This is most commonly used when working with GitHub-type repositories. In this case, it might not be necessary, nor is it as easy to do within the DST environment. So for now, the best approach is to just coordinate and communicate with each other while pushing and pulling to the `main` branch.

9. Protocol

Protocol writing. It is expected that the first output from each publication will be an analysis protocol. There are two broad types of protocols: overall ones for a given work package and specific ones for a given project within (or across) a work package.

> ! Important
>
> Protocols are published to Zenodo. The publishing process will be handled by the Seedcase tool zendo, which makes the process much easier and sometimes can be set up to be done automatically.

9.1 Work package protocols

These work package protocols are (generally) kept in stand-alone repositories, so that they can be directly built into a website and automatically uploaded to Zenodo whenever changes are made. Generally, it's best to have someone from the management group create the repository for the protocol, as its better to set it up using the Template Website from Seedcase that requires some admin settings after creating the repository.

These protocols are bigger and broader in scope, as they represent the overall activity within a work package. They also will have more people working on them, so writing them will be done through pull requests, where at least one other person reviews and approves the changes. In this way, any text on the `main` branch (and built onto the website) will already have been approved and is "publishable" in the current state.

Sections within work package protocols may be more diverse compared to protocols for research projects, which have a template protocol file included with instructions and suggestions.

9.2 Research project protocols

Each research project is created from our template. Within the template contains a file that lists all the sections that should ideally be filled out for a protocol for the research project. When a research project is created from the template, the protocol file is kept in `docs/protocol/protocol.qmd`. See Figure 9.1 for an overview of the protocol writing process.

When you start working on your research project, begin by filling out the protocol file. Inside of it contains instructions and descriptions of different sections typically found in a research project's protocol. Protocols **are not a strict, "binding"** set of things you will do in your project. They are meant to help you as the researcher focus your research question(s) and hypotheses, define your analysis plan, and provide you with a rough roadmap to your final research publication.

Ideally you will want to create a separate Git branch to create your protocol so that you can create a pull request where others can more easily review and approve it. After you've finished writing a protocol and made a pull request, notify someone (who will be coauthor on the project) to review and approve it.

Once the protocol has been reviewed and approved, someone on the management group will publish it on Zenodo for you to obtain a Digital Object Identifier (DOI). This provides a clear record connecting the hypotheses and anticipated analytic approaches with the final analysis of the data.

Any changes to the protocol as the project continues should be updated on Zenodo. If there are any changes, make sure to notify someone on the management group to update the Zenodo deposit and create a new record.

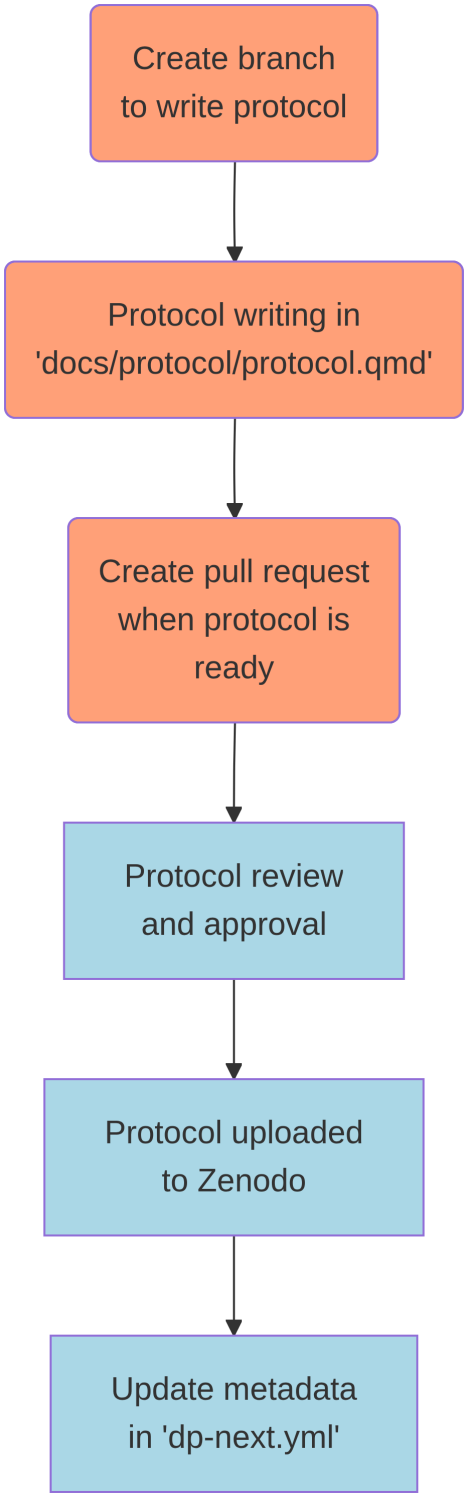


Figure 9.1: Protocol development steps to follow for a research project. The rounded boxes in orange are tasks you are responsible for doing. The rectangular ones in light blue are for coauthors or the management to do.

> **Tip**
>
> If you desire (though it is not required), you could publish a stable version of the protocol as a preprint on online preprint servers such as arXiv or medRxiv. Keep in that you would be responsible for uploading it and keeping it updated. Because publishing to Zenodo is done semi-automatically by someone in the management group, you don't need to do the upload yourself for Zenodo.

9.3 Acknowledgement Statement

Any publication, dissemination, or communication that comes from or is related to DP-Next should mention the following acknowledgement statements.

For all publications:

> The DP-Next project was funded by a Steno National Collaborative Grant from the Novo Nordisk Foundation (Grant number NNF24OC008916). Additional funding was obtained from all Steno Diabetes Centers.

Additionally for publications involving data collected through OPEN, which is used for collecting survey data from e.g. the electronic questionnaires used in Work Package 3 (questionnaire study and the clinical study):

> Electronic questionnaires were distributed using the Open Patient data Explorative Network (OPEN) from Odense University Hospital, Region of Southern Denmark (open.rsyd.dk).

10. Publishing

As detailed in the original project description to the Novo Nordisk Foundation (https://dp-next.github.io/) and Section 12 of the collaboration agreement, we aim to work, collaborate, and publish following the principles of open and reproducible science.

Briefly, the collaboration agreement states that the Steering Committee is responsible for outlining and implementing the project's publication strategy. The main aims of this strategy are:

- To jointly publish positive, negative, and inconclusive results in international peer-reviewed scientific journals.
- To work according to open and reproducible research practices.

To achieve the "open" part of the aim, we'll publish scientific output in journals as Open Access publications, with a strong preference for:

- The Green Open Access format (meaning to publish as a pre-print in an open archive such as Zenodo or bioRxiv).
- Publication agreements that do not require the transfer of copyright to publishers.
- Publishing under a Creative Commons Attribution (CC-BY) license.

> **Note**
>
> In case of any fundamental discrepancy or conflict between this page and the collaboration agreement, the text of the collaboration agreement has precedence. If any substantial deviations from this guide are needed for a particular research project, the authors should justify the deviations and request approval to proceed from the Steering Committee.

So, how do we put this strategy into practice? When a scientific output is ready to be disseminated, either "informally" (in archives like Zenodo or at conferences) or "formally" in journals, in DP-Next we follow a specific path for formal or informal publishing.

10.1 Formally disseminate

This section is only relevant for 'new results', which means any results that have not yet been formally published in a peer-reviewed journal. Results included in a conference abstract or presentation are still considered new results until they are formally published.

When your research project has been reviewed by coauthors, is ready to be submitted to be formally published in an academic journal, you need to inform the management group that your project is ready.

You can do that either by creating an issue on the WP1-Management repo or by sending a message in the "General" Discord channel by pinging the management group with "@admin". An overview of this process is shown below in Figure 10.1.

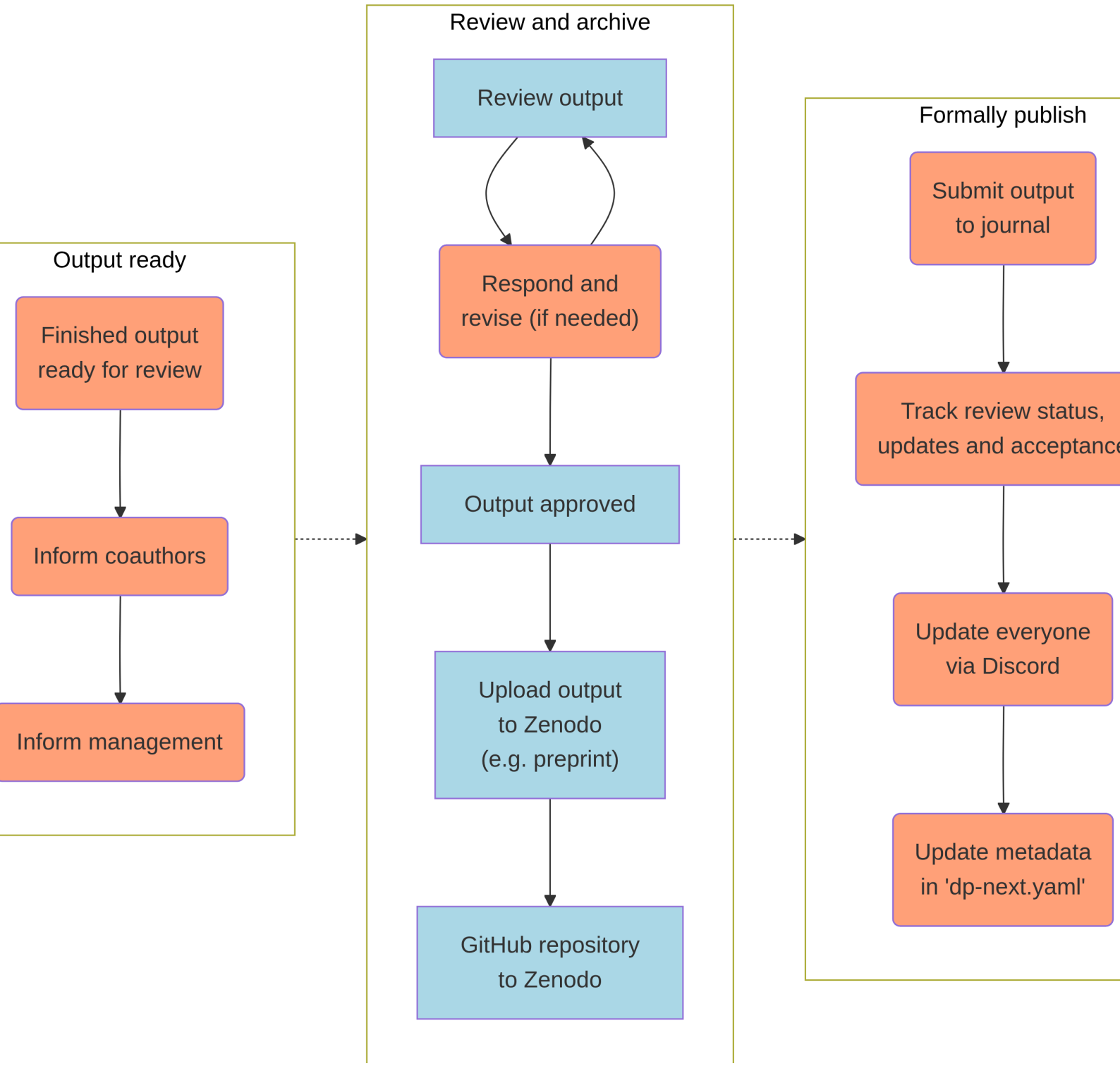


Figure 10.1: Steps to follow for formally publishing scientific output during the life cycle of a DP-Next project. The rounded boxes in orange are tasks you are responsible for doing. The rectangular ones in light blue are for the committee members to do.

You should inform us (the management group) at least two weeks before the intended submission date or deadline. We'll inform you on whether we agree or disagree with the publication as soon as possible and within the two weeks. No response from us can be regarded as agreement and you can continue with the submission.

We'll notify the full Steering Committee, who may assign some of its members to review and approve on the committee's behalf. The Steering Committee will only disagree with the submission of a paper or abstract at this stage if it seems to be premature or is repetition of previous work. Disagreement can also happen if you haven't followed most of the process described in this guide, e.g. no GitHub repository, no approved synopsis, no protocol.

When someone starts reviewing your project, you should respond as quickly as possible to any review comments and to make updates to the project and paper as quickly as possible. After we've given approval, we'll archive the paper as a pre-print to Zenodo and upload the GitHub repository to Zenodo to archive the analysis code.

Once those tasks have been done, you can submit your paper to the journal. It's now your responsibility to track the submission progress, respond to review comments (including making changes to the paper in the GitHub repository), and keep us updated on the status. Once the paper is published, you also need to update the `dp-next.yaml` file in your repository.

10.2 Informally disseminate

If you want to disseminate something that is generally about DP-Next or on formally published results, you don't need approval from the Steering Committee. However, we would still appreciate it if you kept us informed, via the Discord channel or by making an issue on WP1-Management with details about the dissemination activity.

Informal dissemination could include: keynotes, invited talks, press releases, media appearances, podcasts, posts on social media, or outreach activities. In general, these activities must not interfere with or weaken the opportunities for formal academic publication of the study results.

As your project progresses, you'll likely produce various interim results and materials. As far as possible, these should be kept in the project's GitHub repository.

If you want to include preliminary or new results in lectures or seminars that are outside of DP-Next, do so with great care. If you want to present new results in small academic circles, please notify the management group before giving the presentation by creating an issue in the research project's GitHub (or in the WP1-Management repository).

You should ideally only disseminate new results through broader media or outreach activities *after* disseminating the results in as a conference abstract or presentation. In the case of a press release related to the conference, you can prepare the dissemination beforehand under a time embargo that coincides with the academic presentation of results. Before issuing any press release, you should inform the management group via an issue on the (private) WP1-Management repo. You should do this as early as possible. If you need approval in less than 72 hours, send an e-mail to or call one of the members of Steering Committee.

11. Authorship

Authorship publications should reflect a substantial, direct contribution to the work presented in each paper. We aim to use the Vancouver Guidelines as a base and expand on them to reflect nontraditional research contributions. Nontraditional research contributions are activities that contribute in a direct way to the creation of the publication, such as improving data quality, developing software, or improving analysis code.

When you create the project proposal (as in Chapter 7.), you should have an approximate idea of who the authors are and their order in the list. As the project continues, though, this could naturally change. Anyone who has contributed meaningfully or substantially to research project should be given the chance to become coauthor.

All potential coauthors should be involved early in the research project, ideally at the project proposal stage. In addition to the named authors, all papers need to include the group authorship of 'on behalf of the DP-Next investigators', linking to the current list of DP-Next investigators

Keep in mind that author lists and orders aren't fixed at the start and can be amended after project approval by consensus among the authors or by the Steering Committee if the factual contributions differ from those foreseen initially. Ultimately, the Steering Committee has the final authority to decide on and determine the author list of any given paper. If disagreements about authorship cannot be resolved by consensus within the author group, the Steering Committee will make the final decision.

Regardless of named authors, you should make sure to include a detailed contributors list along with their described contributions as per the CRediT roles. At the least, the contributors and CRediT roles should be kept updated in the `dp-next.yaml` file and ideally be mentioned somewhere in the paper or in the supplemental material. The Steering Committee will ensure that such a document is publicly accessible and up to date.

Make sure to also include an acknowledgement to external parties such as funders, institutions, participants, and service providers in papers and other material you present or disseminate elsewhere. The Steering Committee will maintain an updated acknowledgement statement for the DP-Next project in the template-project.

Appendices

Changelog

Since we follow Conventional Commits, we're able to automatically create formal "releases" of the website based on our commit messages. Releases in the context of websites are simply snapshots in time of the website content. We use Cocogitto to be able to automatically create these releases, which uses SemVar as the version numbering scheme, and git-cliff to generate the changelog based on the commit messages.

Because releases are created based on commit messages, a new release is created quite often—sometimes several times in a day. This also means that any individual release will not have many changes within it. Below is a list of the releases we've made so far, along with what was changed within each release.

Commits from bots, like `dependabot` or `pre-commit-ci`, are not included in the changelog.

0.6.1 - 2026-10-27

♻️ Refactor

- Add code to check Git config #35 by `@lwjohnst86` (05c432e)

🚨 Styling

- Update DP-Next Quarto theme #25 by `@lwjohnst86` (ec48676)

0.6.0 - 2026-09-21

✨ Features

- Move publishing content from other repo #18 by `@lwjohnst86` (39a1a9a)
- Add a practicalities section #20 by `@lwjohnst86` (2d1b93d)

♻️ Refactor

- Clarify that WP protocol is made with template-website #21 by `@lwjohnst86` (090d07f)

🚨 Styling

- Update Quarto theme #19 by `@lwjohnst86` (ef62456)

0.5.0 - 2026-09-19

✨ Features

- Add protocol writing section #16 by `@lwjohnst86` (0da6734)

0.4.0 - 2026-09-19

✨ Features

- Move over project management doc #14 by `@lwjohnst86` (31cd09e)

0.3.0 - 2026-09-19

✨ Features

- Move meetings chapter over from wp1-ros #12 by `@lwjohnst86` (a9b3827)

👷 CI/CD

- Comment out broken Quarto render check (for now) #5 by `@lwjohnst86` (a1ad7d0)

❤️ New contributors

- `@pre-commit-ci[bot]` started making automated contributions
- `@dependabot[bot]` started making automated contributions

0.2.0 - 2026-08-30

✨ Features

- Move authorships section over from `wp1-ros` #4 by `@lwjohnst86` (ce029c1)

💄 Styling

- Don't number appendices #1 by `@lwjohnst86` (e876d30)

0.1.1 - 2026-08-30

🐛 Fixes

- Remove `.` before `filename` in code chunks by `@lwjohnst86` (116392c)

👷 CI/CD

- Fix malformed workflow by `@lwjohnst86` (278fce3)

0.1.0 - 2026-08-30

✨ Features

- Add initial Quarto config file and index files by `@lwjohnst86` (1b3dcea)
- Move over section on starting a project by `@lwjohnst86` (e7c19e6)
- Move "roles" chapter into this guide by `@lwjohnst86` (29b4021)
- Move and update the DST setup guide by `@lwjohnst86` (d5e8640)

🐛 Fixes

- Add link to Turing Way website by `@lwjohnst86` (a5f2e85)

♻️ Refactor

- Rename to `overview` from `general` by `@lwjohnst86` (1a71f31)

📝 Documentation

- Minor edits to community health files by `@lwjohnst86` (41f4afd)

💄 Styling

- Add DP-Next Quarto theme by `@lwjohnst86` (fb45ae0)

🤦 Miscellaneous

- Start of guide repo by `@lwjohnst86` (8feebcc)
- Switch to use `dp-next-theme` in justfile by `@lwjohnst86` (06fce32)
- Update and run pre-commit hooks by `@lwjohnst86` (f937cf2)
- Exclude `_book/` and `_extensions/` from TODO listing by `@lwjohnst86` (ae8107a)
- Ignore Quarto `*_files/` by `@lwjohnst86` (102e2ed)

❤️ New contributors

- `@github-actions[bot]` started making automated contributions
- `@lwjohnst86` made their first contribution

Contributing

:bug: Issues and bugs

The easiest way to contribute is to report issues or bugs that you might find while reading through the website. You can do this by creating a new issue on our GitHub repository.

:pencil2: Adding or modifying content

If you would like to contribute content, please check out our guidebook for more specific details on how we work and develop. It is a regularly evolving document, so is at various states of completion.

To contribute to `guide`, you first need to install uv and justfile. We use uv and justfile to manage our project, such as to run checks and test the template. Both the uv and justfile websites have a more detailed guide on using uv, but below are some simple instructions to get you started.

It's easiest to first install uv and then install justfile with uv. Once you've installed uv, install justfile by running:

```
uv tool install rust-just
```

We keep all our development workflows in the `justfile`, so you can explore it to see what commands are available. To see a list of commands available, run:

```
just
```

As you contribute, make sure your changes will pass our tests by opening a terminal so that the working directory is the root of this project's repository and running:

```
just run-all
```

When committing changes, please try to follow Conventional Commits as Git messages. Using this convention allows us to be able to automatically create a release based on the commit message by using Cocogitto. If you don't use Conventional Commits when making a commit, we will revise the pull request title to follow that format. That's because we squash merge when merging pull requests, so all other commits in the pull request will be squashed into one commit.

:file_folder: Explanation of files and folders

This is a description of some of the files in this repository.

- `.copier-answers.yml`: Contains the answers you gave when copying the project from the template. **You should not modify this file directly.**
- `.github/`: Contains GitHub-specific files, such as issue and pull request templates, workflows, dependabot configuration, pull request templates, and a CODEOWNERS file.
- `_quarto.yml`: Quarto configuration file for the website, including settings for the website, such as the theme, navigation, and other options.
- `_metadata.yml`: Quarto metadata file for the website, including information about the project, such as the titles and GitHub names.
- `.gitignore`: This ignore file tells Git which files to not track. Unless you know what you are doing, it's best to not touch this file.
- `.pre-commit-config.yaml`: Pre-commit configuration file for managing and running checks before each commit.
- `.config/`: Contains configuration files for various tools used in the project, such as:
  - `typos.toml`: typos spell checker configuration file.
  - `rumdl.toml` and `panache.toml`: rumdl and Panache configuration file for formatting Markdown files in the project.

  - `cog.toml`: Cocogitto configuration file for managing versions.
  - `cliff.toml`: git-cliff configuration file for creating the changelog.
- `.editorconfig`: Editor configuration file for EditorConfig to maintain consistent coding styles across different editors and IDEs.
- `.zenodo.json`: Structured citation metadata for your project when archived on Zenodo. This is used to add the metadata to Zenodo when a GitHub release has been uploaded to Zenodo.
- `justfile`: `just` configuration file for scripting project tasks.
- `CHANGELOG.md`: Changelog file for tracking changes in the project.
- `CONTRIBUTING.md`: Guidelines for contributing to the project.

Contributor Code of Conduct

As contributors and maintainers of this project, we pledge to respect all people who contribute through reporting issues, posting suggestions, updating any material, submitting pull requests, and other activities.

We are committed to making participation in this project a harassment-free experience for everyone, regardless of level of experience, gender, gender identity and expression, sexual orientation, disability, personal appearance, body size, race, ethnicity, age, or religion.

Examples of unacceptable behavior by participants include the use of sexual language or imagery, derogatory comments or personal attacks, trolling, public or private harassment, insults, or other unprofessional conduct.

Project maintainers have the right and responsibility to remove, edit, or reject comments, commits, code, wiki edits, issues, and other contributions that are not aligned to this Code of Conduct. Project maintainers who do not follow the Code of Conduct may be removed from the project team.

Instances of abusive, harassing, or otherwise unacceptable behavior may be reported by opening an issue or contacting one or more of the project maintainers.

This Code of Conduct is adapted from the Contributor Covenant (https://www.contributor-covenant.org), version 1.0.0, available at https://www.contributor-covenant.org/version/1/0/0/